AI-Based Universal JEE Question Generation System

Compact System Design & Architecture

Objective: Generate unlimited mathematically-valid JEE/NEET question variants automatically from a single expert-authored image.

1. Problem Statement

Manual creation of JEE/NEET questions is slow and difficult to scale. Random numerical replacement often produces mathematically invalid or visually inconsistent questions.

Problem	Impact
Manual Authoring	30–60 minutes per question
Constraint Violations	Invalid equations and diagrams
Low Scalability	Hard to generate large datasets
Visual Inconsistency	Poor diagram quality

2. System Overview

1. Question Upload
Original image input

2. GPT-4 Vision
Semantic JSON extraction

3. SymPy
Constraint validation

4. Qwen2-VL
Layout-aware edit planning

5. Renderer
Diagram & equation generation

6. Output
Variant question images

3. Semantic JSON Generation

GPT-4 Vision converts uploaded question images into structured semantic JSON templates containing variables, formulas, constraints, and solution metadata.


```
{ "subject": "Physics", "chapter": "Projectile Motion", "variables": [ { "name": "theta", "range": [30,80],
"sampled_value": 45 } ], "constraints": [ "0 < theta < 90" ] }
```


4. Constraint Validation Engine

Constraint	Purpose
Pythagoras	Triangle validity
Snell's Law	Optics correctness
Positive Resistance	Circuit validity
Angle Bounds	Physically meaningful values

5. Controlled Image Regeneration

Stage	Description
Qwen2-VL	Generates edit instructions
Region Mapping	Maps values to image coordinates
Matplotlib/SVG	Diagram rendering
LaTeX/mathtext	Equation rendering
Pillow	Final image composition

6. Final Unified Architecture

Component	Technology	Role
Semantic Extraction	GPT-4 Vision	Image $\rightarrow$ JSON
Constraint Engine	SymPy	Validation
Layout Understanding	Qwen2-VL	Edit planning
Diagram Rendering	Matplotlib/SVG	Diagram generation
Equation Rendering	LaTeX/mathtext	Equation generation
Image Compositing	Pillow	Final PNG creation

7. Conclusion

The proposed architecture combines multimodal reasoning, symbolic mathematics, layout-aware editing, and deterministic rendering to generate scalable, mathematically-valid JEE/NEET question variants.